**Solution:** Deviation from assumed mean

| Sl. No | (X) | d=X-A (A=57) | $d^2$ |
|---|---|---|---|
| 1 | 43 | -14 | 196 |
| 2 | 48 | -9 | 81 |
| 3 | 65 | 8 | 64 |
| 4 | 57 | 0 | 0 |
| 5 | 31 | -26 | 676 |
| 6 | 60 | 3 | 9 |
| 7 | 37 | -20 | 400 |
| 8 | 48 | -9 | 81 |
| 9 | 78 | 21 | 441 |
| 10 | 59 | 2 | 4 |
| N=10 | | $\Sigma d = -44$ | $\Sigma d^2=1952$ |

$$\sigma = \sqrt{\frac{\Sigma d^2}{n} - \left(\frac{\Sigma d}{n}\right)^2} = \sqrt{\frac{1952}{10} - \left(\frac{-44}{10}\right)^2}$$

$$= \sqrt{195.2\text{-}19.36} = \sqrt{175.84} = 13.26,$$

**Answer = 13.26**

## 12.12 Correlation (ϒ)

### Introduction

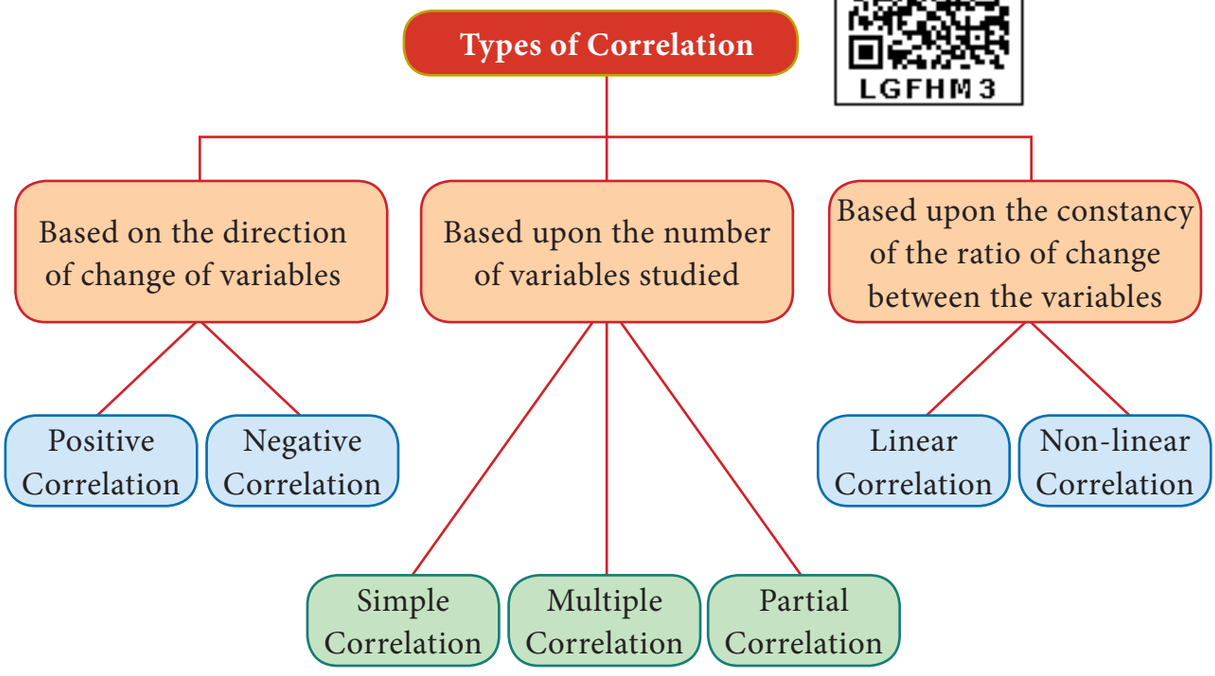
Correlation is a statistical device that helps to analyse the covariation of two or more variables. Sir Francis Galton, is responsible for the calculation of correlation coefficient.

### Types of Correlation


Karl Pearson

Correlation is classified in several different ways. Three of the most important ways of classifying correlation are:

LGFHM3




### Type I: Based on the direction of change of variables

Correlation is classified into two types as Positive correlation and Negative Correlation based on the direction of change of the variables.



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